CSC4093: Neural Networks and Deep Learning

Programming Assignment 01
Tweet Classification using LSTM Networks

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1. Introduction

This report covers the implementation of two LSTM based models for classifying tweets as personal health mentions or non-personal health mentions. The PHM dataset was used for training and evaluation. A Stacked LSTM and a Bidirectional LSTM were built and compared in terms of accuracy, precision, recall and F1 score. The main challenge with this kind of data is the informal nature of tweets, where short phrases and missing context make classification harder than typical text data.

2. Dataset

The dataset contains tweets labelled as 1 (personal health mention) or 0 (non-personal). It was split into 9,991 training samples and 3,331 test samples. A further 20% of the training data was held out for validation during model training.

Split	Count
Train	9,991
Test	3,331

3. Methodology

3.1 Preprocessing

Tweets were cleaned by lowercasing, removing URLs, mentions and non-alphabetic characters. Stop words were removed but negation words like 'not' and 'never' were kept on purpose since dropping them would flip the meaning of many health related phrases. Single character tokens were also discarded.

3.2 Tokenization

A Keras Tokenizer was fitted on training data only, producing a vocabulary of 11,136 tokens. Sequences were padded to a length of 15 which is the 90th percentile of training lengths. This was chosen instead of the maximum length to avoid excessive padding from a few very long tweets.

3.3 Stacked LSTM

The first model uses two LSTM layers in sequence. The first layer has 64 units and passes its full output to the second layer which has 32 units. A Dense layer with 32 units and sigmoid output handles classification. The first layer picks up lower level word patterns and the second layer builds on those to understand the overall meaning of the tweet.

3.4 Bidirectional LSTM

The second model wraps each LSTM layer in a Bidirectional layer so the sequence is read both forward and backward. This gives the model more context at each word position which helps when meaning depends on what comes after a word, not just before it. The layer structure is otherwise the same as Model 1.

3.5 Overfitting Control

In early runs, training accuracy went above 93% while validation accuracy stayed around 80%, which was a clear sign of overfitting. SpatialDropout1D was added after the embedding layer, dropout of 0.5 and recurrent dropout of 0.3 were applied inside the LSTM layers, and L2 regularization of 0.0001 was added to the weights. EarlyStopping with patience 5 was used to stop training when validation loss stopped improving, and ReduceLROnPlateau halved the learning rate when progress stalled.

4. Results

4.1 Performance Summary

Metric	Stacked LSTM	Bidirectional LSTM
Accuracy	82.38%	82.2%
Precision	0.8179	0.8161
Recall	0.8238	0.822
F1 Score	0.8177	0.8164
Correct / Total	2744 / 3331	2738 / 3331
Epochs	8	8

4.2 Stacked LSTM Training History

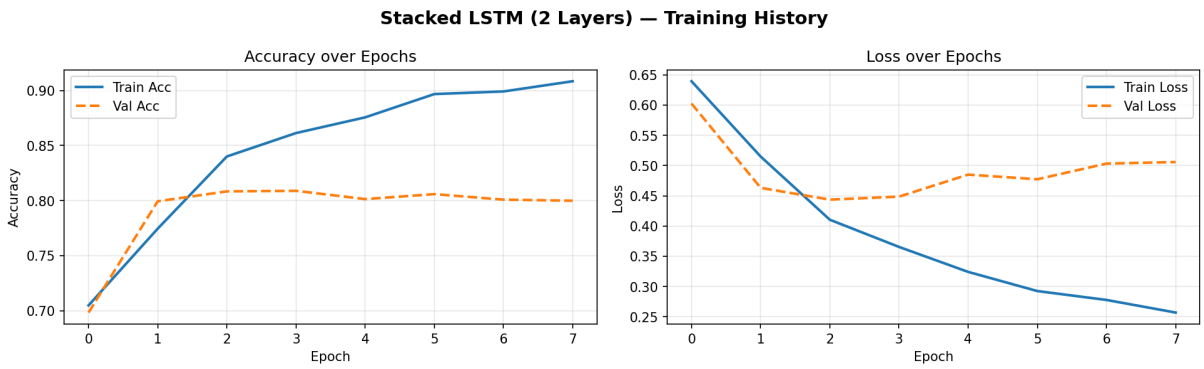

Figure 1: Accuracy and loss over epochs for the Stacked LSTM.

4.3 Bidirectional LSTM Training History

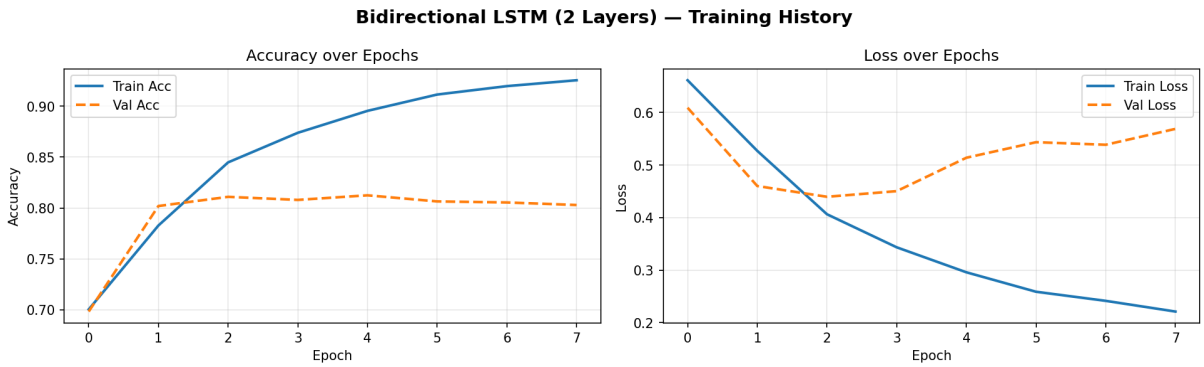

Figure 2: Accuracy and loss over epochs for the Bidirectional LSTM.

4.4 Confusion Matrices

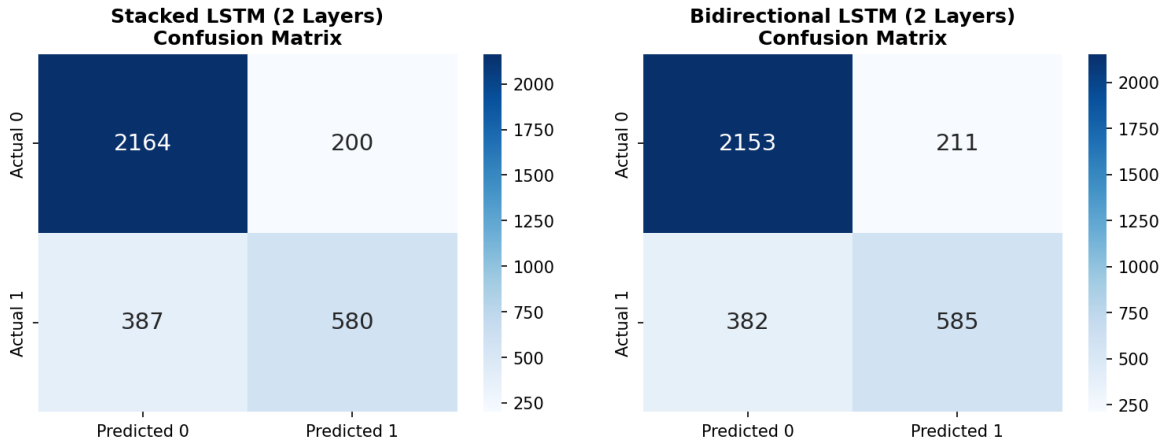

Figure 3: Confusion matrix for Stacked LSTM (left) and Bidirectional LSTM (right).

5. Discussion

The Stacked LSTM performed better overall, reaching 82.38% accuracy compared to 82.2% for the Bidirectional LSTM. The difference of 0.18 points is consistent across all metrics which suggests it is not just random variation. Reading tweets in both directions seems to help the model understand context better, especially when the key word appears mid or end of the sentence.

The training curves show that the regularization worked well. In earlier attempts without dropout and L2, the validation loss was going up while training loss kept falling. After adding the regularization stack both curves moved together. EarlyStopping made sure the final weights came from the best validation epoch rather than the last one.

Keeping negation words during preprocessing also made a difference. Something like 'not feeling well' would become 'feeling well' after standard stop word removal which completely flips the meaning. The 90th percentile for sequence length also helped avoid the model learning from excessive padding in short tweets.

6. Conclusion

Both models managed to classify personal health mentions reasonably well. The Stacked LSTM came out ahead with 82.38% accuracy and an F1 of 0.8177. The regularization choices kept overfitting under control and the preprocessing decisions helped preserve meaning in the tweet text. Using pre-trained Twitter embeddings like GloVe or a model like BioBERT would probably push the accuracy higher, but within the scope of this assignment the results are reasonable and the models generalise well on unseen data.

Appendix: Classification Reports

A.1 Stacked LSTM

	precision	recall	f1-score	support
0	0.85	0.92	0.88	2364
1	0.74	0.60	0.66	967
accuracy			0.82	3331
macro avg	0.80	0.76	0.77	3331
weighted avg	0.82	0.82	0.82	3331

A.2 Bidirectional LSTM

	precision	recall	f1-score	support
0	0.85	0.91	0.88	2364
1	0.73	0.60	0.66	967
accuracy			0.82	3331
macro avg	0.79	0.76	0.77	3331
weighted avg	0.82	0.82	0.82	3331